The aims of this session:

To explore and understand quantum strategies for time series data

Problem
Quantum Technology
Qubits
Variational Quantum Algorithms
Quantum Neural Networks
VQA jako szeregi Fouriera
Case Studies
Summary

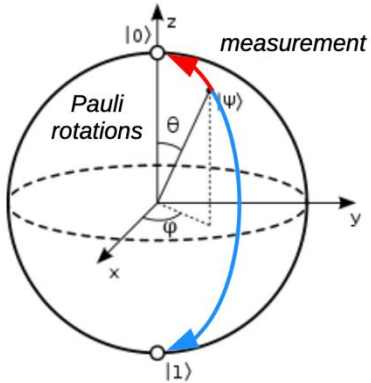
Time under the Qubit's control: Quantum Analysis of Time Series

Czas to to, co sprawia, że wszystko nie dzieje się na raz

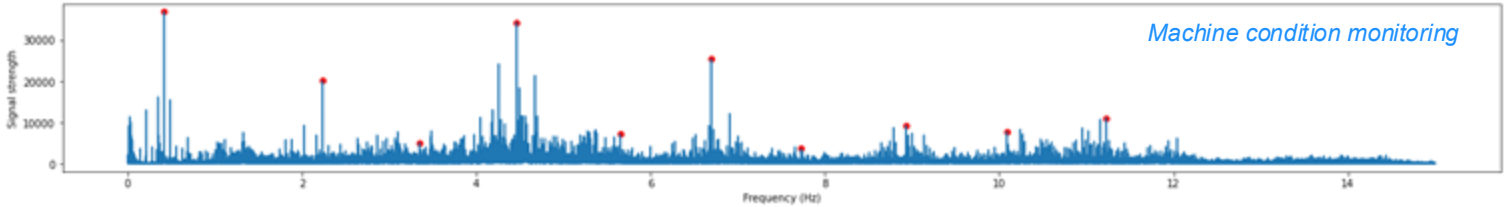
Sebastian Zając
SGH Szkoła Główna Handlowa w Warszawie

Jacob L. Cybulski
Enquanted, Melbourne, Australia

12-14.10.2025, MZBO 2025

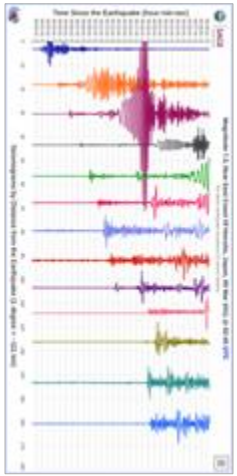


Problem

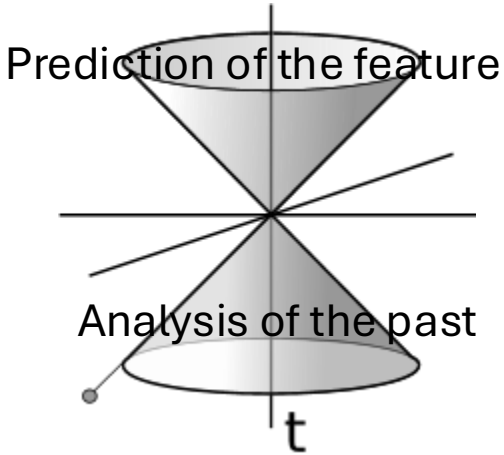


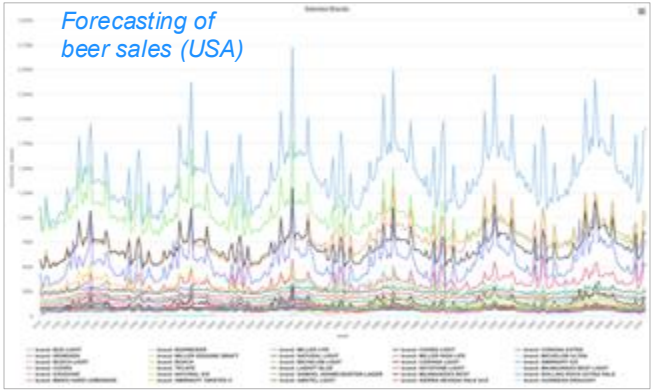
Machine condition monitoring

EEG analysis



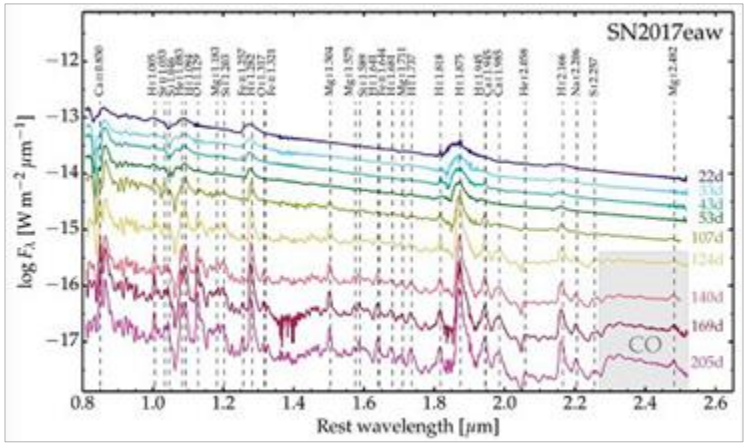
Earthquakes





Forecasting of beer sales (USA)

Astronomical observations

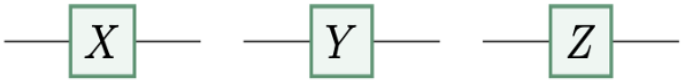


Can **quantum machine learning** assist detection of complex patterns in time series and signals from the preceding data sequences? How?

What is
Quantum Technology

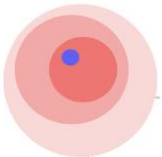
Quantum computing *and*
Quantum information science
study of the information processing tasks that can be accomplished using quantum mechanical systems (Nielsen and Chuang, 2010)

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$



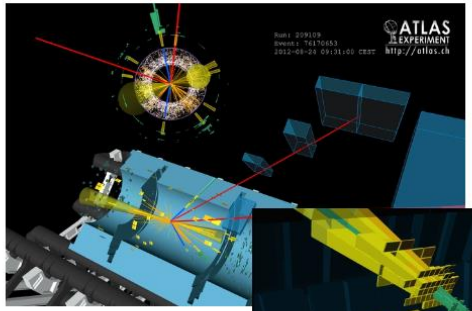
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Quantum finance
Quantum chemistry
Quantum optimisation
Quantum machine learning
problem-solving with quantum technology



What is a Quantum Computer?

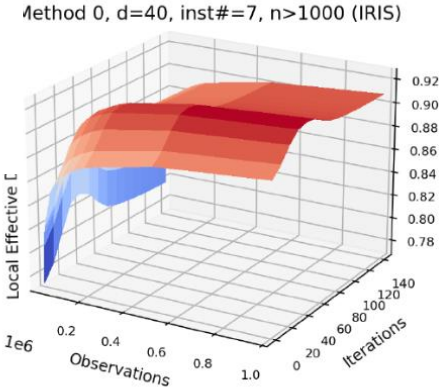
It is a device which directly applies the principles of quantum mechanics to perform computational tasks

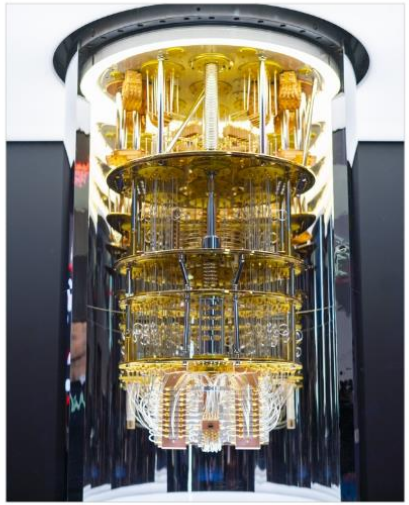


Quantum mechanics
area of science dealing with the behaviour of matter and light on the atomic and subatomic scale (Britannica.com, 2020)

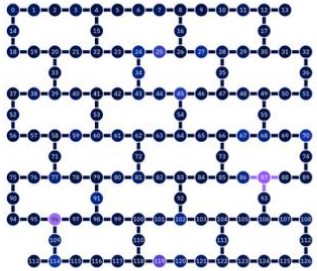
CERN
Atlas particle detector
Large Hadron Collider

There are other quantum technologies e.g. quantum sensing



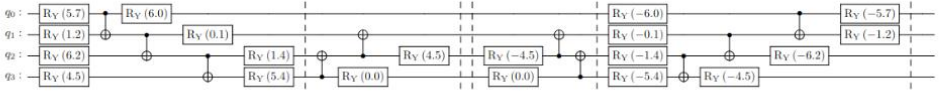


IBM superconducting quantum machine (127 qubits on cloud)



Quantum engineering
building quantum devices

Cybulski & Nguyen
Barren Plateaus in QNNs and VQAs



Recently in the news ...

Quantum Machines

(2024)

Univ of Sci and Tech of China with Shanghai Inst of Microsystem and Info Tech (*Jiuzhang 3* - Photonic)

Again achieved quantum supremacy

Gaussian Boson Sampling

SpinQ
(N. Magnetic Resonance)

IQM / VTT
(Superconducting)

IBM (Superconducting)

D-Wave
(Quantum Annealing)

PASQAL
(Neutral Atoms)

They all use:

Qubits
the fundamental models of quantum information and its processing

Quantum circuits
models of computation, involving qubits and operations on them

Quantum Brilliance
(Diamond)

Microsoft Azure Quantum

AWS Braket

(Platforms)

Providers
- Quantinuum
- IonQ
- QCI
- Rigetti
- Pasqal
- 1QBit
- Microsoft QIO
- Toshiba ...

Providers
- IonQ
- Rigetti
- OQC
- Xanadu
- D-Wave
- QuEra ...

Xanadu (Photonic)

Google
(Superconducting)

Qubits

in scientific terms

In practice qubits involve *elementary particles*, such as *ions*, *photons*, single *atoms*, *electrons*, even *defects in diamonds*; and, their behaviour is governed by Physics (Nature / Universe)

A qubit *represents a state* of such a particle, e.g. an electron spin, which can be up or down, (written formally as $|\uparrow\rangle$ and $|\downarrow\rangle$ or $|0\rangle$ and $|1\rangle$), which are called the *basis states*

It is possible to change the state of a qubit with certain predetermined *operations*, such as rotation or reversing the position of the qubit state

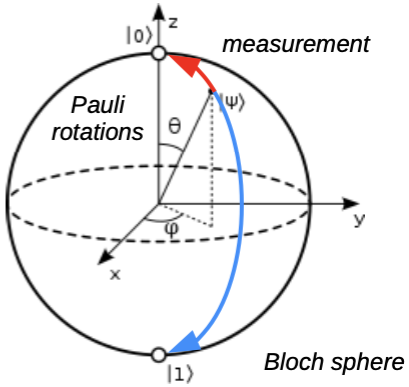
Afterwards, the qubit is in a state of *superposition*, or a combination, of its basis states up and down

The superposition state is the actual state of elementary particles, not its math description

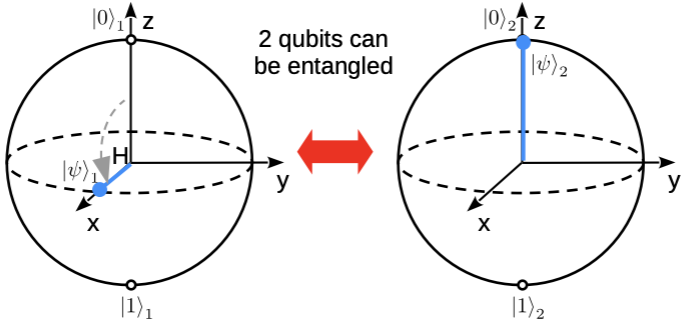
It is impossible to determine the qubit's state without its *measurement*

Qubit measurement returns only the basis state that is "likely" to be closest to its superposition state, which also destroys the qubit state

The outcome of measurement is precise but probabilistic



Qubits can be *entangled*, then they start behaving as a unit with a common complex state, until they are measured or until some external factor (*noise*) destroys their entanglement



What makes quantum computers special?

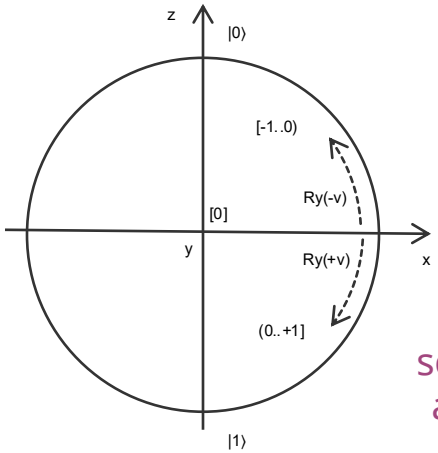
Qubit *superposition* (parallel choices) and *entanglement* (exponential combination of choices and their filtering), as well as *measurement* (collapse of choices and randomness), is what gives quantum computers their immense computational power allowing some problems to be solved in minutes rather than 1000s of years!

Input encoding / embedding
for QNN processing

Generally, QAE input and output are an unrestricted collection of real values (floats), guiding our selection of data encoding methods.

We rejected the following encoding methods:

Basis encoding, with qubits acting as bits in the encoded number (logical / int) to be processed later in the circuit.

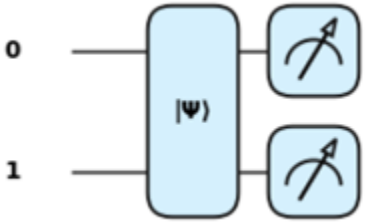


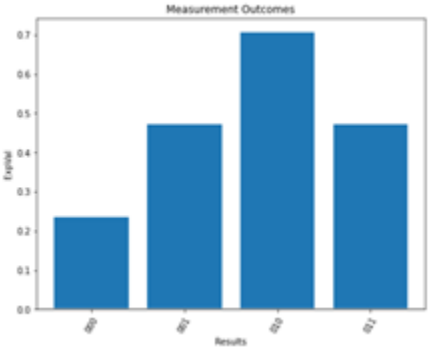
Angle encoding suits QAE design, with input values represented as Qubit state rotations (float).

In our experiments, we used angle encoding relative to $|+\rangle$ state, with values $\in [-1, 1]$ scaled to a range $[0, \pi]$, and coded as rotations up (<0) or down (≥0).

Amplitude encoding is probably the least understood, however, it is one of the most useful encoding schemes.

It embeds input as a circuit state normally measured on output, i.e. , each data point is encoded as the expectation value of a multi-qubit measurement (int/*float)*.





Example: data encoded as ψ was normalised vector [1/8, 2/8, 3/8, 2/8]. The measurement reflects the input data proportions.

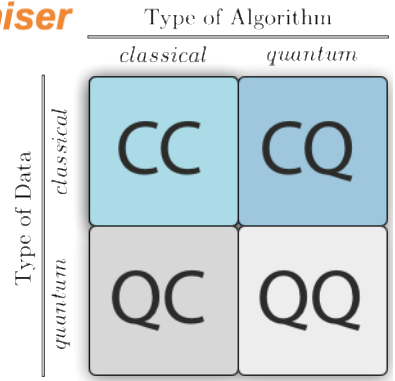
Variational Quantum Algorithm

VQA is an ***iterative process***

VQA uses ***cost/loss function*** and ***optimiser***

VQA has ***difficulties***:

- The problem at hand
- Large circuits with many parameters
- Complex measurement strategy
- Unsupervised learning
- Emergence of barren plateaus



A typical VQA process

The ansatz parameters are initialised to some values, e.g. zero or random

The feature map parameters are bound to the new input data

The parameter values are used to create a new circuit

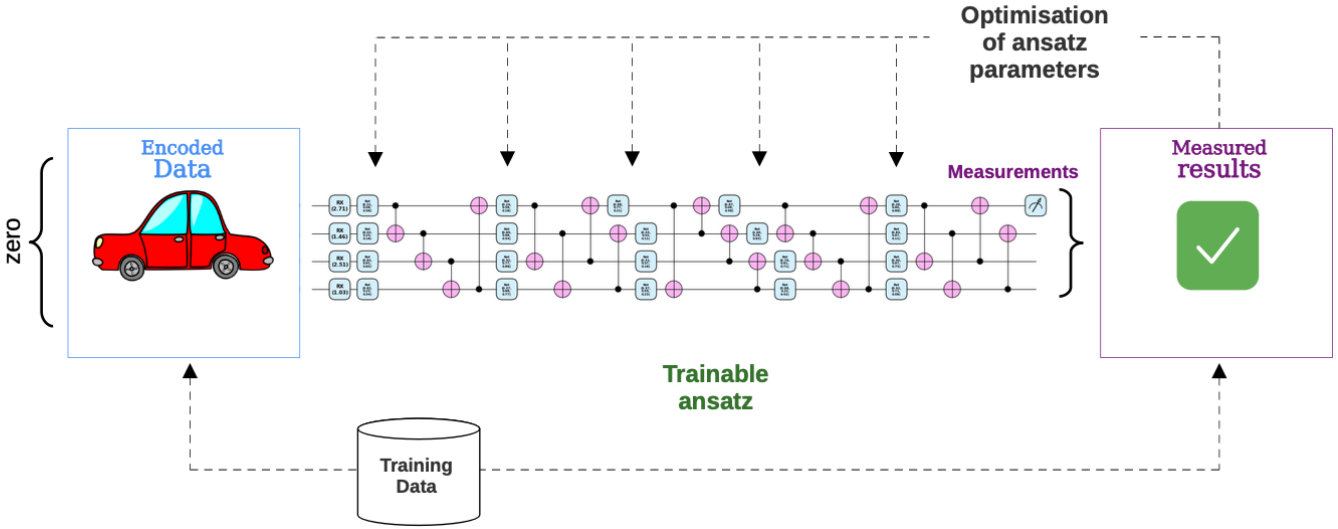
The circuit is executed

The circuit quantum state is then measured

Cost function is applied to measurement results and expected values

The cost of difference is calculated

Based on the difference and previous parameters the new parameters values are proposed



Quantum Neural Networks

QNNs are implemented as parameterized quantum circuits consisting of:

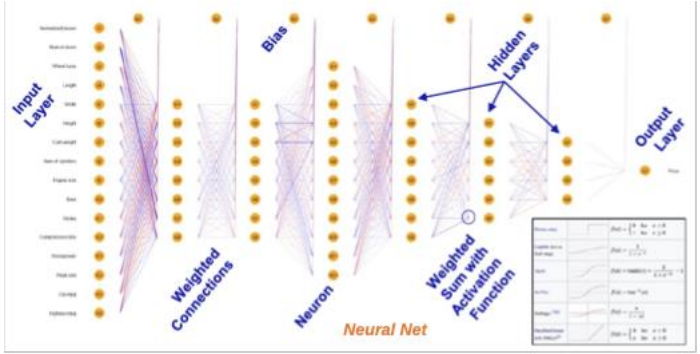
- The feature map: a layer that encodes classical data.

- The parameters: consists of several layers and, similarly to a classical neural network, is responsible for interconnecting the layers.

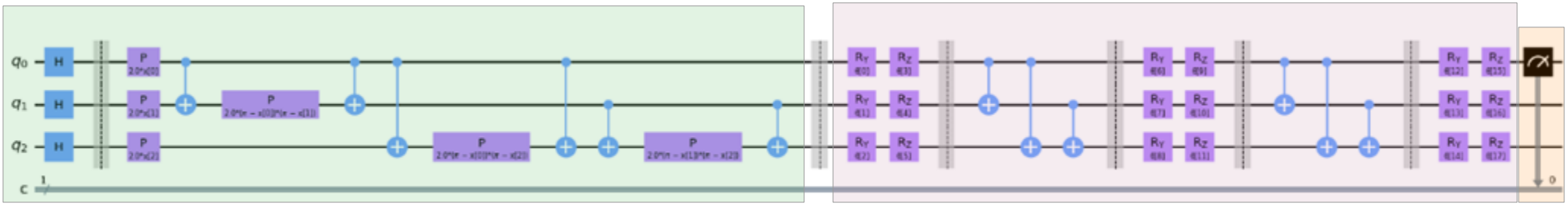
- Measurement of the qubit states: interpreting the obtained results.

„Classical" Neural Nets for Time Series Analysis

- trends, seasonality for stationary TS
- Recurrent Neural Networks
- Long Short-Term Memory
- Gated Recurrent Units



Feature map *Parameters* *Measurement*



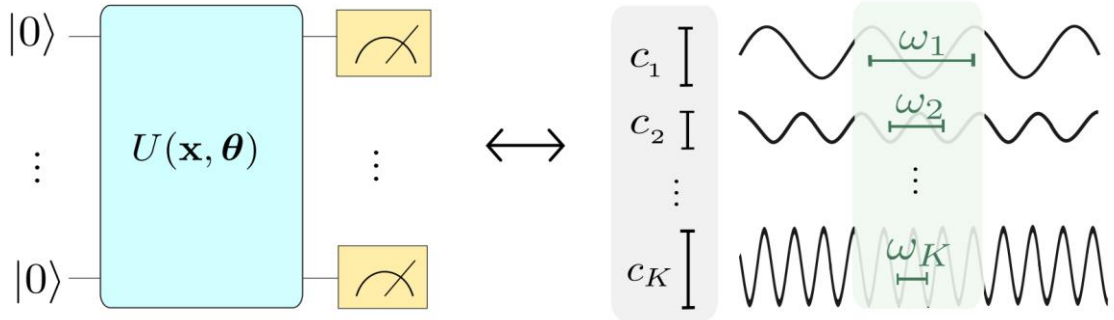
VQA as a Fourier Series

The class of functions a quantum model can represent is characterized by the model's ability to control the frequencies and coefficients of its Fourier series.

$$f_{\boldsymbol{\theta}}(x) = \langle 0|U^{\dagger}(x,\boldsymbol{\theta})MU(x,\boldsymbol{\theta})|0\rangle$$

$$f_{\boldsymbol{\theta}}(x) = \sum_{\omega\in\Omega} c_{\omega}(\boldsymbol{\theta})\, e^{i\omega x}.$$

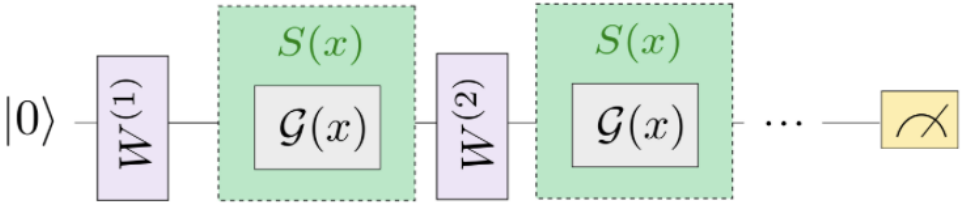
Where M is an observabel
and $U(x,\theta)$ is a variational quantum circuit that encodes a **data input** x and depends on a **set of parameters** θ.

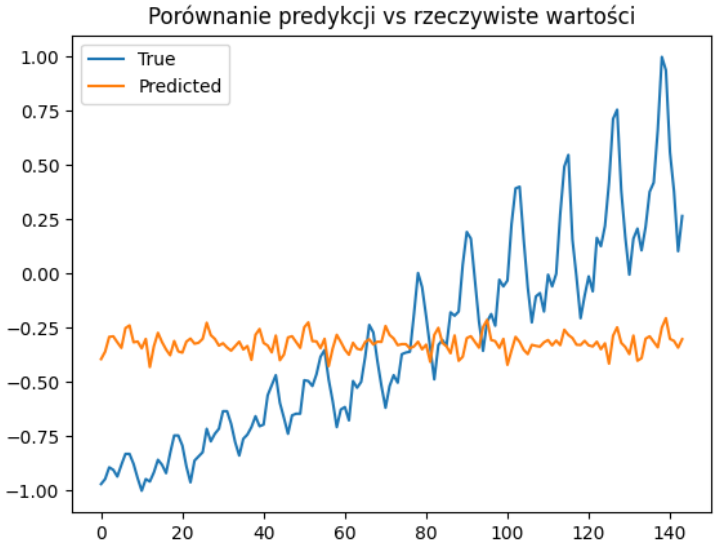


The effect of data encoding on the expressive power of variational quantum machine learning models by Schuld, et all (2020)

Fourier series with serial Pauli-rotation encoding - 1 qubit

$$f_{\boldsymbol{\theta}}(x) = \langle 0|U^{\dagger}(x,\boldsymbol{\theta})MU(x,\boldsymbol{\theta})|0\rangle$$

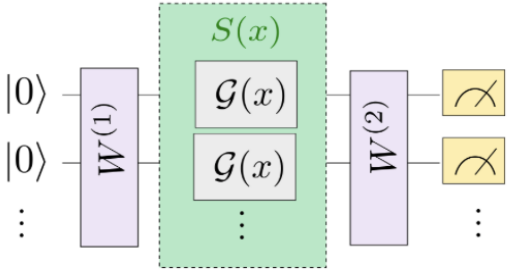


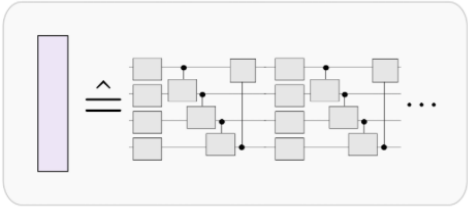


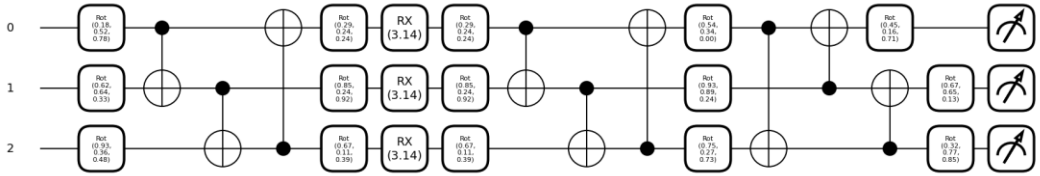
The effect of data encoding on the expressive power of variational quantum machine learning models by Schuld, et all (2020)

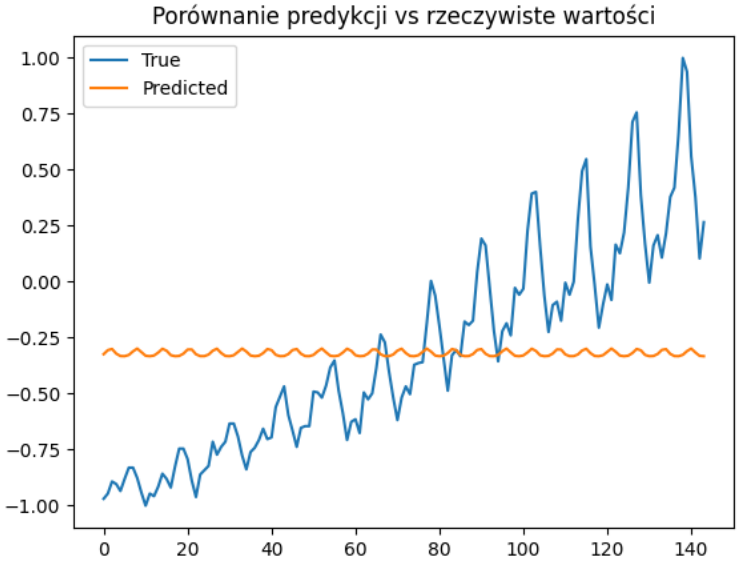
Fourier series – parallel quantum model 3 qubits

$$f_{\boldsymbol{\theta}}(x) = \langle 0|U^{\dagger}(x,\boldsymbol{\theta})MU(x,\boldsymbol{\theta})|0\rangle$$









The effect of data encoding on the expressive power of variational quantum machine learning models by Schuld, et all (2020)

Directed acyclic graph

- It is easy to imagine more interesting ways we could combine **quantum** and **classical** ingredients into a larger and more **complex hybrid computation**. In general, one can understand these subroutines as *classical* and *quantum nodes*.

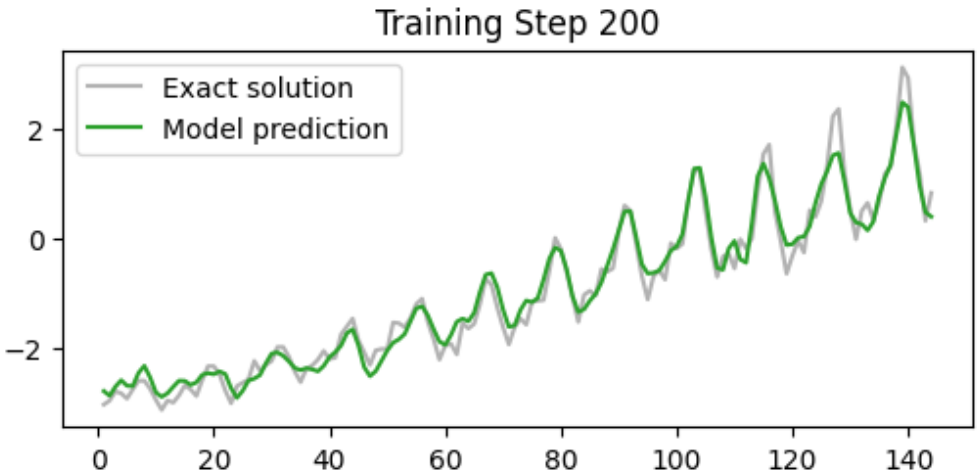
- Each node in the graph can be either classical or quantum, and quantum nodes running on different devices (e.g., a qubit and a CV device) can be combined in the same computation.

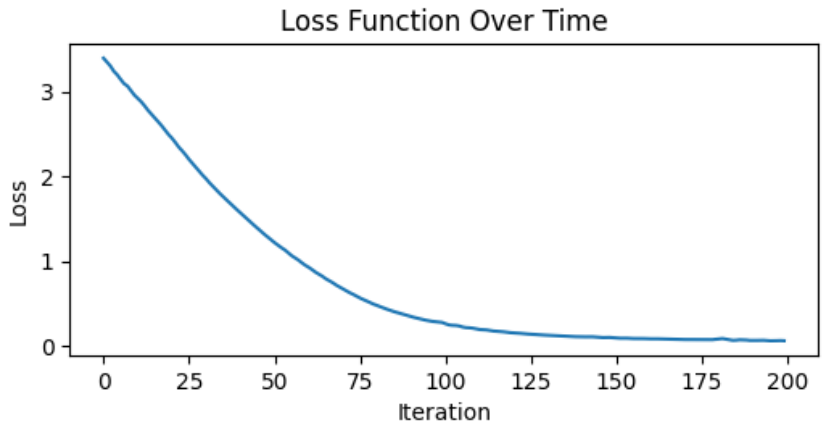
```python
class QN(torch.nn.Module):
    '''Classical -> Quantum -> Classical'''

    def __init__(self, n_input: int, n_output: int, quanutm_layer):
        super().__init__()

        self.layers = torch.nn.Sequential(
            torch.nn.Linear(n_input, n_qubits),
            quanutm_layer,
            torch.nn.Linear(n_qubits, n_output)
        )


    def forward(self, x):
        return  self.layers(x)
```





Quantum Autoencoder (for Time Series)

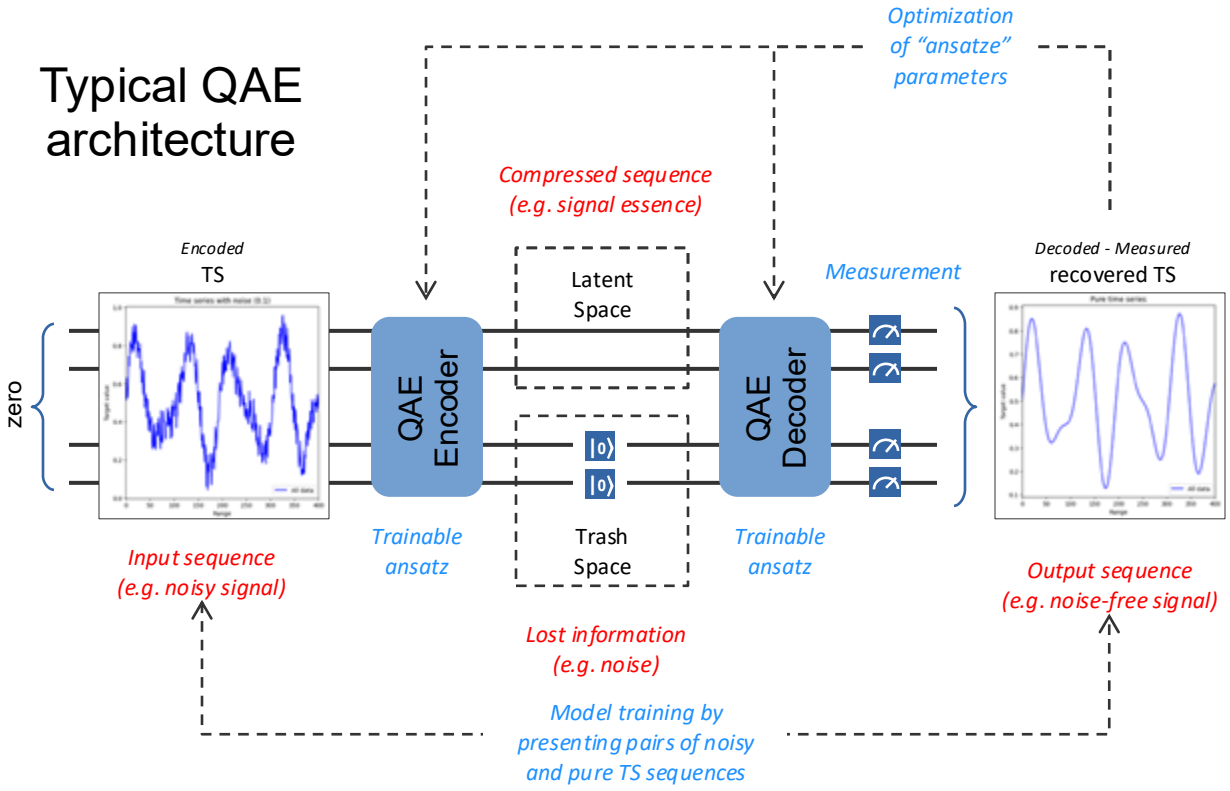
There are a few applications of QML methods to time-series analysis, TS applications of quantum AE (QAE) are even fewer

QAEs have the potential to deal with highly complex noise and anomaly patterns

Training of QAEs is difficult, due to:

- Potentially many features (e.g. TSs) (lots of qubits and/or parameters)
- Complex measurement strategies
- Unsupervised learning (we do not know what is noise)
- Possibility of barren plateaus

Typical QAE architecture



In QAE development, the key concerns include:
overall model architecture, data encoding and decoding, ansatz design and its parameters optimisation strategy

Simple problems can be solved with pure quantum methods
Complex issues require hybrid quantum-classical methods

Thank you!